

2024

B.Tech. 3rd Semester End-Term Examination

SOLID MECHANICS

(New Regulation)

Full Marks – 70

Time – Three hours

The figures in the margin indicate full marks
for the questions.

Answer question No. 1 and any *four* from the rest.

1. Choose the correct option 1 × 10
- (i) Tensile Strain in a steel rod is the ratio between:
- (a) Change in length to original volume
 - (b) Increase in length to original length .
 - (c) Decrease in length to original length
 - (d) Change in length to diameter of rod
- (ii) A steel rod of 125 cm long and 2 cm diameter is subjected to an axial load of 18 KN. The stress in MPa is:
- (a) 0.72
 - (b) 4.73
 - (c) 57.32
 - (d) 0.95
- (iii) The percentage elongation and the percentage reduction in area depends upon
- (a) Ductility of the material
 - (b) Tensile strength of the material
 - (c) Hardness of the material
 - (d) Toughness of the material
- (iv) A beam is said to be of uniform strength, if _____
- (a) B.M. is same throughout the beam
 - (b) Shear stress is the same through the beam
 - (c) Deflection is the same throughout the beam
 - (d) Bending stress is the same at every section along its longitudinal axis

[Turn over

- (v) What is the expression of the bending equation (with usual notations)?
- (a) $M/R = \sigma/y = E/I$ (b) $M/Y = \sigma/R = E/I$
- (c) $M/I = \sigma/R = E/y$ (d) $M/I = \sigma/y = E/R$
- (vi) As per Euler's theory, the longest effective length of a column is obtained when
- (a) Both the ends are hinged
- (b) One end fixed while the other end is free
- (c) Both the ends are fixed
- (d) One end fixed while the other end is hinged
- (vii) A Column of 3 m length is effectively held in position at both the ends and restrained against rotation at one end only. Its effective length will be:
- (a) 3 m (b) 2.1 m
- (c) 1.5 m (d) 6 m
- (viii) A thin cylinder shell of diameter 'd' is subjected to an internal pressure 'p'. The ratio of longitudinal stress to the hoop stress is:
- (a) 1 (b) 2
- (c) $\frac{1}{2}$ (d) $\frac{3}{4}$
- (ix) The hoop stress in a thin cylindrical shell is:
- (a) Circumferential tensile stress
- (b) Tensile stress
- (c) Compressive stress
- (d) None of the above
- (x) The modulus of rigidity of a circular shaft is 25 units and the polar moment of inertia 10 units. The torsional rigidity will be:
- (a) 2.5 units (b) 250 units
- (c) 0.4 units (d) 125 units

2. (a) Define: Shear stress and Bending stress. 2

(b) Deduce the following expression with usual notations: 6

$$E = 3K \left(1 - \frac{2}{m} \right)$$

- (c) A steel bar of varying cross-section is subjected to some forces as shown in the figure 1. Find the total changes in length. Assume the modulus of elasticity as 210000 MPa.

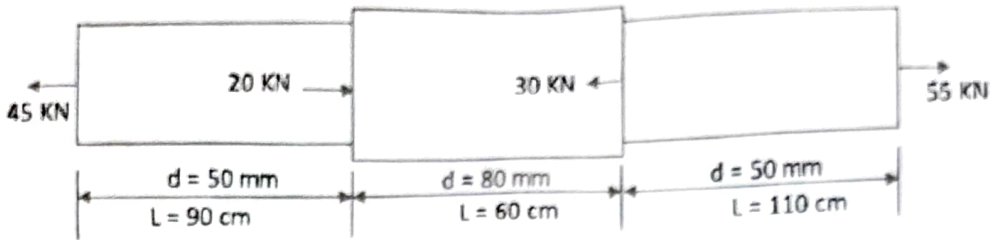


Figure 1

3. (a) Draw and discuss the shear stress distribution of a rectangular section of width b and depth d . 5
- (b) A rectangular beam 250 mm deep and 100 mm wide is simply supported over a span of 5 m. If the shear force at the section of the beam is 50 kN, then
- (i) Calculate the average and maximum shear stress of the section. 5 + 5
- (ii) Find out the shear stress at a distance of 25 mm above the neutral axis of the beam section.
4. (a) Using the usual notations, derive the expression for Euler's Cripling load of a column with both the ends fixed. 8

$$P = \frac{4\pi^2 EI}{L^2}$$

- (b) A hollow cylindrical cast iron column is 4 m long with one end fixed and the other is free. Determine the minimum diameter of the column if it has to carry a safe load of 220 kN with the F.O.S as 2.8. Assume the internal diameter as 0.75 times the external diameter. Assume the ultimate crushing strength as 500 N/mm² and Rankine's constant as 1/1600. 7
5. (a) What are the different kinds of stresses acting on a cylinder? 3
- (b) Derive the expression for the circumferential stress developed in a cylindrical shell of diameter " d " and wall thickness as " t " which is subjected to " p " internal fluid pressure. 5

- (c) A thin cylinder of 1.5 m internal diameter contains a fluid at an internal pressure of 2.2 N/mm^2 . Determine the wall thickness if the Hoop's stress and longitudinal stress is not to exceed 40 N/mm^2 and 32 N/mm^2 respectively.

4 + 3

If the joint efficiencies along circumferential and axial directions are 70% and 40% respectively, what will be the required wall thickness of the cylinder?

6. (a) Derive the expression for the torque (T) developed in a solid circular shaft of radius "R" subjected to shear stress " τ ". 8
- (b) Determine the Torque transmitted for a hollow circular shaft subjected to the shear stress of 50 N/mm^2 . The external diameter of the shaft is 140 mm and the wall thickness is 15 mm. 5 + 2

If the shaft rotates at 20 revolutions every minute, find the power generated by the shaft.

7. (a) Draw a neat sketch for a double riveted lap joint. 3
- (b) State the step-wise procedure for the determination of the normal and tangential stresses using Mohr's circle method. 5
- (c) A water pipe 80 cm in diameter contains water at an internal pressure head of 100 m. If the specific weight of water is 9810 N/m^3 , find the thickness of the metal used to make the pipe for conveying water. Assume the permissible stress in the pipe as 20 N/mm^2 . 7

If the fluid pressure is increased by 20%, what will be the required thickness?